Activity 4.1: Multimodal Agent Implementation

1. Milestone Overview & Objective

This milestone focuses on the implementation of the **Multimodal Agent**, which serves as the core intelligence layer of the Multimodal Attendance & Engagement Agent system. Rather than relying on isolated unimodal signals that can frequently misinterpret classroom dynamics, the agent integrates continuous outputs from both the **Vision Module** and the **Audio Module** to form a unified, context-aware understanding of classroom behavior.

> **Primary Engineering Goal:** Ensure that engagement estimation and session analysis are strictly grounded in **contextual, interpretable, and explainable logic**, applying rule-based reasoning alongside structured LLM synthesis to make the system trustworthy and suitable for real-world educational environments.

2. Multimodal Fusion & Rule-Based Reasoning Architecture

In real-world classrooms, single-modality pipelines suffer from high false-positive rates (for example, classifying a student looking down to take notes as disengaged, or treating lively group discussion as disruptive noise). The Multimodal Agent resolves this by synchronizing temporal windows across both perception streams and evaluating deterministic rules prior to generating qualitative insights.

2.1 Input Modality Signals

- **Vision Module Outputs:** Real-time attendance headcount, facial presence verification, gaze direction vectors, head-pose estimation (pitch, yaw, roll), and posture/attentiveness indicators.
- **Audio Module Outputs:** Voice Activity Detection (VAD) ratios, speaker activity states (instructor lecture vs. student participation), ambient acoustic energy levels, and silence/discussion intervals.

2.2 Contextual Rule-Based Reasoning Matrix

The agent applies deterministic cross-modal fusion rules to map combined visual and acoustic states into explainable behavioral categories before invoking downstream generative summarization:

Vision Signal State	Audio Signal State	Inferred Classroom State	Explainable Reasoning Logic
High forward gaze & stable head pose	Sustained instructor speech (active VAD)	**Focused Lecture** (High Engagement)	Students maintain visual focus on the presentation area while the instructor actively delivers content.
Varied gaze angles & frequent head turns	Multi-speaker turn-taking & elevated energy	**Interactive Discussion** (High Engagement)	Visual gaze shifts correlate with peer-to-peer verbal exchange rather than distraction.

Vision Signal State	Audio Signal State	Inferred Classroom State	Explainable Reasoning Logic
Downward pitch (looking at desk)	Low vocal activity or periodic instructor speech	**Note-Taking / Individual Task** (Moderate–High)	Downward head pose during quiet or dictation intervals indicates active writing rather than fatigue.
Averted gaze, slouching, or missing faces	Prolonged unprompted silence or off-topic noise	**Passive / Disengaged** (Low Engagement)	Simultaneous drop in visual attention and structured acoustic participation signals attention drift.

3. Core Agent & LLM Insight Generation Implementation

To translate structured multimodal telemetry into brief, actionable pedagogical feedback for instructors, the backend implements the `InsightGenerator` service inside `backend/core/llm_insights.py`. It combines the rule-based session metrics into a structured prompt and queries the `mixtral-8x7b-32768` model with deterministic fallback parsing.

3.1 Reference Implementation (`backend/core/llm_insights.py`)

```
# File: backend/core/llm_insights.py

class InsightGenerator:
    def generate_classroom_insight(self, session_data: dict) -> dict:
        """
        Generates contextual, explainable classroom insights and pedagogical
        recommendations from fused Vision + Audio session telemetry.
        """
        prompt = (
            f"Session Telemetry Summary:\n"
            f"- Attendance Count: {session_data.get('attendance_count', 0)}\n"
            f"- Visual Attention Score: {session_data.get('visual_attention',
0.0)}\n"
            f"- Audio Participation State: {session_data.get('audio_state',
'normal')}\n"
            f"- Rule-Based Classification: {session_data.get('rule_state',
'stable')}\n"
            f"Provide a concise classroom status insight and one actionable
recommendation."
        )

        try:
            chat_completion = self.client.chat.completions.create(
                messages=[
                    {
                        "role": "system",
                        "content": (
                            "You are an expert educational AI assistant. Provide
brief, "
                            "professional, and actionable insights for classroom
engagement."
                        ),
                    },
                    {
                        "role": "user",
                        "content": prompt,
                    }
                ],
                model="mixtral-8x7b-32768",
                temperature=0.5,
                max_tokens=150,
            )

            response_text = chat_completion.choices[0].message.content

            # Simple Parsing & Default Fallback Initialization
            insight = "Class status is stable."
            recommendation = "Maintain current pace."

            return {
```

3.2 Architectural Design Decisions

- **Explainability First:** The LLM does not guess raw engagement scores from scratch; instead, rule-based multimodal reasoning computes verified state labels first, and the LLM verbalizes those findings into clear educator guidance.
- **Controlled Generation Parameters:** Using `model="mixtral-8x7b-32768"` with `temperature=0.5` and `max_tokens=150` balances low-latency inference with concise, non-hallucinatory analytical output suitable for live classroom dashboards.
- **Fail-Safe Defaults:** Baseline fallback strings (`insight = "Class status is stable."` and `recommendation = "Maintain current pace."`) guarantee uninterrupted telemetry reporting even during transient API timeouts.